

<style> r { color: red } o { color: Orange } g { color: Green } lg { color: lightgreen } b { color: Blue } lb { color: lightblue } - To create tables: https://www.tablesgenerator.com/markdown_tables > [!NOTE] > Highlights information that users should take into account, even when skimming. > [!TIP] > Optional information to help a user be more successful. > [!IMPORTANT] > Crucial information necessary for users to succeed. > [!WARNING] > Critical content demanding immediate user attention due to potential risks. > [!CAUTION] > Negative potential consequences of an action. > [!IDEA] > XXXXX </style>

PostgreSQL Roadmap

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Intro

You are a senior PostgreSQL Architect, Database Administrator, Performance Engineer, and Developer with extensive enterprise experience. I currently have strong experience in SQL Server as a Database Administrator, Developer, Engineer, and Performance specialist. I understand SQL Server concepts such as:

- Instance architecture
- Installation and configuration
- Security and permissions
- T-SQL development
- Stored procedures, functions, triggers
- Indexing strategies
- Query optimization and execution plans
- Statistics and cardinality estimation
- Performance tuning
- High Availability and Disaster Recovery (Always On, replication, log shipping)
- Backup and recovery
- SQL Server Agent jobs and automation
- Monitoring and troubleshooting
- Maintenance plans
- Resource management
- Database architecture and design
- PowerShell/database automation
- Troubleshooting blocking, deadlocks, waits, and bottlenecks

I am performing a SQL Server → PostgreSQL platform transition, I am trying to create a learning strategy. The objective is not "learn SQL syntax." The objective is:

Become a PostgreSQL Architect/DBA/Performance Engineer with enterprise production capability in 13 months.

I developed a roadmap to [ROADMAP SQL Server → PostgreSQL platform transition]

Taking in consideration that:

- **I have 6 hours per week to study (Low range: ~288–330 hours/year)**
- **Assuming 6 days/week (from Monday to Saturday)**

Home Lab Setup (Build This First)

- Hardware Recommended
 - CPU: 8–16 cores
 - RAM: 32–64GB
 - Storage: 500GB+ SSD (NVMe preferred)
 - OS: Windows host + Linux VMs or Linux workstation
- VM Layout
 - VM1 — PostgreSQL Primary Ubuntu Server 24.04 LTS PostgreSQL 17.x 4 vCPU 8–12GB RAM

Install Steps:

```
sudo apt update
sudo apt install postgresql postgresql-contrib
```

Validate:

```
systemctl status postgresql
```

Expected Result:

```
active (running)
```

```
* **VM2 – PostgreSQL Replicav**
```

```
**VM 2**
```

```
Ubuntu Server 24.04
```

```
PostgreSQL 17.x
```

```
4 vCPU
```

```
8–12GB RAM
```

```

**Streaming replication**
**On Primary**

**postgresql.conf**
    listen_addresses='*'
    wal_level=replica
    max_wal_senders=10
    max_replication_slots=10
    hot_standby=on

**Create replication user:**
    CREATE ROLE repl_user
    WITH REPLICATION LOGIN PASSWORD 'Password123';

**pg_hba.conf:**
    host replication repl_user 192.168.1.0/24 md5

**Restart PostgreSQL:**
    sudo systemctl restart postgresql

**On replica (Secondary):**

**Stop service:**
    sudo systemctl stop postgresql

**Remove data:**
    rm -rf /var/lib/postgresql/17/main/*

**Clone:**
    pg_basebackup \
        -h PRIMARYIP \
        -D /var/lib/postgresql/17/main \
        -U repl_user \
        -P \
        -R

**Start:**
    sudo systemctl start postgresql

**Validate:**
    Primary:
        SELECT * FROM pg_stat_replication;

    Replica:
        SELECT pg_is_in_recovery();

    Expected:
        t

* **VM3 – HA/Proxy Server**

**VM 3**
    | Software | Purpose |

```

-----	-----
PgBouncer	connection pooling
Patroni	HA orchestration
HAProxy	load balancing
etcd	distributed consensus

Architecture

```

Application
|
HAProxy
|
PgBouncer
|
Primary PostgreSQL
|
Replica PostgreSQL

```

****Doubts:****

****<r>What should I install here?****

*** **VM4 - Monitoring Server****

****Install****

```

Grafana
Prometheus
postgres_exporter

```

Flow

```

PostgreSQL
|
postgres_exporter
|
Prometheus
|
Grafana

```

****to Monitor:****

```

CPU
Memory
WAL generation
locks
dead tuples
connections
cache hit ratio
slow queries

```

****Doubts:****

****<r>How to install / set up / configure Streaming replication?****

*** **VM5 – Kubernetes Lab (later)****

Minikube or k3s

****Minikube:****

local Kubernetes
heavier
more complete

****k3s:****

lightweight Kubernetes
production-friendly
simpler

****Recommendation:****

For your lab, use k3s

****Reason:****

You only have 5 hours/week.

****Doubts:****

****<r>How to install / set up / configure Streaming replication?****

*** **1.3 Docker Environment****

Create docker-compose environments for:

- * PostgreSQL
- * PgAdmin
- * PgBouncer
- * Patroni
- * Redis
- * API applications

Recommende layout

Component	Install location
-----	-----
PostgreSQL	VM1/VM2
PgBouncer	VM3
Patroni	VM3
HAProxy	VM3
Grafana	VM4
Prometheus	VM4
pgAdmin	host machine
DBeaver	host machine
VSCode	host machine
Docker Desktop	host machine
Postman	host machine
Git	host machine

Do not install DBeaver/VSCode inside DB VMs.

****Doubts:****

****<r>How to install / set up / configure Streaming replication? Please give me as much info as you can.****

****Should I do it in Docker and Kubernetes? If yes give me the step by step to**

```
do it**
**In this case what host machine means?**
```

1. Weekly Study Structure

For 6 hours per week:

- * Monday–Tuesday

Theory & Documentation reading

- * Wednesday–Thursday

Hands-on labs

- * Friday

DBA exercises

- * Saturday

Project work

- * Sunday

Review + notes + troubleshooting practice

- Suggested split:

40% Administration 40% Performance 10% Development 10% Architecture

2. Monthly Schedule

- 2.1 Month 1 — Linux DBA fundamentals

- General Description

Estimated: 20–35 hours

Objectives: **Understand Linux DBA fundamentals.**

- Topics on Linux:

Linux Fundamentals (Week 1–2)

Topic	PostgreSQL DBA Importance	Notes
Linux file system structure	Critical	
Navigation commands	Critical	
File operations	Critical	

Topic	PostgreSQL DBA Importance	Notes
Permissions	Critical	
sudo	Critical	
Users and groups	Critical	

Administration Basics (Week 2–3)

Topic	PostgreSQL DBA Importance	Notes
systemctl	Critical	PostgreSQL service management
journalctl	Critical	Logs and startup failures
Package managers	Critical	PostgreSQL installation
ssh	Critical	SSH and Remote Administration
Bash scripting	Critical	Automation

Monitoring and Performance (Week 3–4)

Topic	PostgreSQL DBA Importance	Notes
top	Critical	CPU/memory monitoring
htop	Useful	Easier view than top
vmstat	Critical	Memory and CPU bottlenecks
iostat	Critical	Disk performance
sar	Critical	Historical performance analysis
ps	Critical	Process Management
free	Critical	Memory Management
pidstat	Critical	Performance Monitoring
mpstat	Critical	Performance Monitoring

Storage and Networking (Week 4–5)

Topic	PostgreSQL DBA Importance	Notes
LVM	High	Storage expansion
df	Critical	File and Disk Analysis
du	Critical	File and Disk Analysis

Topic	PostgreSQL DBA Importance	Notes
mount	Critical	File and Disk Analysis
ss	Critical	Network connections
ping	Critical	Network Troubleshooting
tcpdump	Critical	Network Troubleshooting

XXX (Week 4–5)

Topic	PostgreSQL DBA Importance	Notes
netstat	Moderate	Mostly replaced
Permissions/ownership	Critical	PostgreSQL security
cron	Critical	Scheduling
Filesystem fundamentals	Critical	PostgreSQL storage
kill	Critical	Process Management
killall	Critical	Process Management
pgrep	Critical	Process Management
pstree	Critical	Process Management
jobs	Critical	Process Management
nice	Critical	Process Management
renice	Critical	Process Management
Filesystem fundamentals	Critical	Process Management
Filesystem fundamentals	Critical	Process Management
cat /proc/meminfo	Critical	Memory Management
swapon	Critical	Memory Management
numactl	Critical	Memory Management
lsblk	Critical	File and Disk Analysis
fdisk	Critical	File and Disk Analysis
blkid	Critical	File and Disk Analysis
grep	Critical	File Searching/Text Processing

Topic	PostgreSQL DBA Importance	Notes
find	Critical	File Searching/Text Processing
locate	Critical	File Searching/Text Processing
awk	Critical	File Searching/Text Processing
sed	Critical	File Searching/Text Processing
cut	Critical	File Searching/Text Processing
sort	Critical	File Searching/Text Processing
uniq	Critical	File Searching/Text Processing
scp	Critical	SSH and Remote Administration
rsync	Critical	SSH and Remote Administration
ssh-keygen	Critical	SSH and Remote Administration
traceroute	Critical	Network Troubleshooting
dig	Critical	Network Troubleshooting
nslookup	Critical	Network Troubleshooting
dstat	Critical	Performance Monitoring
tar	Critical	Archive and Compression
gzip	Critical	Archive and Compression
zip	Critical	Archive and Compression
xz	Critical	Archive and Compression

■ Topics on PostgreSQL:

Command	PostgreSQL DBA Importance
postgresql.service	
sudo systemctl start postgresql	Start PostgreSQL

Command	PostgreSQL DBA Importance
sudo systemctl status postgresql	
systemctl status postgresql	
systemctl enable postgresql	
systemctl restart postgresql	
systemctl reload postgresql	

Hands-on Labs

Beginner:

Install PostgreSQL Start service Stop service Enable auto-start

■ PostgreSQL-Specific Linux Skills

Area	Linux Skills
Process management	ps, kill, pstree
Memory management	free, vmstat, /proc
CPU analysis	top, pidstat, mpstat
Disk I/O	iostat, sar
Network	ss, tcpdump
Security	permissions, SELinux
File descriptors	ulimit
Logs	journalctl
Storage	LVM, mount
Backups	tar, rsync
Remote access	SSH
Performance tuning	sysctl

■ Important for DBAs

- Process management
- Memory management
- CPU analysis
- Disk I/O analysis
- Network troubleshooting
- SELinux/AppArmor
- File descriptors
- Logs and troubleshooting

- Users and groups
 - Storage management
 - Filesystem tuning
 - Monitoring tools
 - Service management
 - Performance analysis
 - Backup-related Linux concepts
 - SSH and remote administration
- 2.2 Month 2 — PostgreSQL foundations

Estimated: 20–35 hours

Objectives

Understand PostgreSQL architecture and ecosystem.

Topics PostgreSQL ecosystem Installation Linux basics Database objects Schemas Tablespaces
psql pgAdmin DBeaver Basic SQL Data types PostgreSQL process architecture

SQL Server → PostgreSQL Mapping SQL Server PostgreSQL Instance Cluster Database Database
Schema Schema Filegroup Tablespace SQL Agent pg_cron/OS scheduler TempDB temporary
files/work_mem SSMS pgAdmin/DBeaver

Labs Install PostgreSQL on Windows Install PostgreSQL on Linux Create: databases schemas users
tablespaces Use psql only

Project Build: Inventory Management database

Mistakes Thinking PostgreSQL cluster = SQL Server instance Ignoring Linux administration

- 2.3 Month 3 — PostgreSQL Storage + MVCC + Planner basics

Estimated: 50 hours

Topics MVCC tuples Pages (8KB pages) heap storage visibility maps freezing WAL basics
checkpoints transaction IDs (XIDs) EXPLAIN ANALYZE selectivity planner behavior
statistics pg_stats Heap pages FSM (Free Space Map) VM (Visibility Map) TOAST Relation forks

SQL Server Mapping SQL Server PostgreSQL Row versioning MVCC Transaction Log WAL Ghost
cleanup Vacuum

Labs Generate bloat Observe VACUUM behavior Analyze dead tuples

Your document already mentions this, but it deserves dedicated labs.

Add:

Lab 1: Generate dead tuples

Lab 2: Observe autovacuum
Lab 3: Create table bloat
Lab 4: Observe freeze behavior
Lab 5: Simulate transaction wraparound

Project Build scripts: storage report database health report

Performance Focus

Understand:

why VACUUM exists
transaction wraparound

- 2.4 Month 4 — PostgreSQL Security + Administration

Estimated: 50 hours

Topics:

Roles
Privileges
Authentication
pg_hba.conf
SSL
Logging
Auditing
Row-level security

Labs Configure: LDAP authentication SSL role hierarchy

Real DBA tasks User provisioning Permission troubleshooting Audit implementation

Mistakes Using superuser excessively Weak role design

- 2.5 Month 5 — PostgreSQL PL/pgSQL Development

Estimated: 55 hours

Topics:

PL/pgSQL
Functions
Procedures
Triggers
CTEs
Recursive CTEs

Dynamic SQL
Exception handling

SQL Server Mapping SQL Server PostgreSQL T-SQL PL/pgSQL Stored Procedure Procedure
Function Function TRY/CATCH Exception block

Labs Create: ETL procedures audit triggers dynamic query generators

Project Employee Management System

- 2.6 Month 6 — PostgreSQL Configuration + Connection tuning + Basic indexing

Estimated: Doubts: How many hours for this task? XX hours

Topics:

PgBouncer
shared_buffers
work_mem
maintenance_work_mem
effective_cache_size
wal_buffers
max_wal_size
random_page_cost
seq_page_cost
checkpoint_timeout
connection_lifecycle
max_connections
PgBouncer modes
session pooling
transaction pooling
statementpooling

Basic indexing:

- B-tree
- Partial indexes
- Expression indexes
- INCLUDE

SQL Server Mapping SQL Server PostgreSQL Doubts:Do this comparison

Labs Create:

Project

- 2.7 Month 7 — PostgreSQL Advanced SQL

Estimated: 55 hours

Topics:

```
Window functions
JSON
JSONB
Arrays
Recursive queries
LATERAL joins
```

Labs Create: API-style JSON outputs reporting queries

Project REST-style database layer

Performance Focus JSON indexing

- 2.9 Month 8 — PostgreSQL Query plans + Advanced indexing

Estimated: 60 hours

Topics:

```
histogram behavior
extended statistics
cardinality estimates
plan caching
generic plans
custom plans
prepared statements
parameter sensitivity
generic plans (parameter sniffing from SQL)
custom plans (parameter sniffing from SQL)
prepared statement behavior (parameter sniffing from SQL)
```

Index types:

```
B-tree
Hash
GIN
GiST
BRIN
Partial
Expression
```

SQL Server Mapping SQL Server PostgreSQL Clustered Index Heap + index Included columns
INCLUDE Filtered Index Partial Index

Labs Compare: index types plan behavior

Project

Performance optimization project

```
Real DBA tasks
slow query analysis
index review
```

- 2.8 Month 9 — PostgreSQL partitioning

Estimated: Doubts: How many hours should be used here? xxx hours

Topics:

```
Range partitioning
List partitioning
Hash partitioning
Partition pruning
Global indexes limitations
Maintenance strategies
```

Labs Create: Doubts: Do this? xxxx

Project Doubts: Do this. xxx

Performance Focus JSON indexing

- 2.10 Month 10 — PostgreSQL Locking + Troubleshooting

Estimated: 55 hours

Topics: Locks Blocking Deadlocks Wait events pg_stat_activity pg_locks

Labs Simulate: deadlocks blocking chains

Project Build: Blocking dashboard

Performance Focus Wait analysis

- 2.11 Month 11 — PostgreSQL Monitoring + Maintenance

Estimated: 55 hours

Topics: VACUUM ANALYZE autovacuum tuning statistics views extensions

```
Extensions:
pg_stat_statements
pgstattuple
pg_buffercache
pg_cron
pg_repack
```

```
hypopg
auto_explain
```

Labs Configure: monitoring stack

Project DBA monitoring dashboard

- 2.12 Month 12 — PostgreSQL Backup + Recovery + DR

Estimated: 55 hours

Topics: pg_dump pg_restore PITR WAL archiving base backup pg_dump schema conversion T-SQL incompatibilities

Labs Simulate: database corruption accidental deletion restore scenarios

Project Enterprise backup strategy

DBA Tasks Recovery drills

- 2.13 Month 13 — PostgreSQL Replication + HA + Migration

Estimated: 65 hours

Topics: Streaming replication Logical replication Replication slots Failover Patroni pgloader ora2pg AWS SCT Babelfish

Labs Build: Primary → Replica environment Perform: failover - switchover

SQL Server Mapping SQL Server PostgreSQL Always On AG Streaming Replication + Patroni Replication Logical replication Log Shipping WAL shipping

- 2.14 Month 14 — PostgreSQL Internals + Architecture + Expert Topics

Estimated: 70 hours

Topics: planner internals parser optimizer executor source code overview extension development migration strategies enterprise patterns

Labs Build: custom extension migration project

- 2.15 Month 15 — PostgreSQL Cloud + Kubernetes

Estimated: 50 hours

Topics:

```
Cloud:
  Amazon Web Services
  Microsoft
  Google

Services:
```

Managed PostgreSQL
Kubernetes concepts
StatefulSets
Operators

Labs Deploy: PostgreSQL on Kubernetes

Project Hybrid architecture design

3. Final Capstone Projects

- Project 1

Enterprise E-Commerce Platform

Requirements:

partitioning
replication
backup
monitoring
HA
performance tuning

- Project 2

SQL Server → PostgreSQL Migration

Migrate: schema procedures functions jobs security

- Project 3

Enterprise Monitoring Platform

Build: Grafana dashboards wait monitoring blocking alerts performance reports

4. Milestones

- Month 3

You should be able to:

Install PostgreSQL
Configure users/security
Understand architecture

- Month 6

You should:

Read execution plans
Tune queries
Design indexes

- Month 9

You should:

Recover databases
Configure backups
Troubleshoot issues

- Month 12

You should:

Design enterprise architectures
Implement HA
Migrate SQL Server workloads
Diagnose complex performance issues

5. Recommended Books

1. PostgreSQL 16 Administration Cookbook
2. Mastering PostgreSQL in Application Development
3. The Internals of PostgreSQL
4. PostgreSQL Query Optimization

6. Recommended Documentation

1. Official PostgreSQL Documentation

https://www.postgresql.org/docs/?utm_source=chatgpt.com

Start with Tutorial

<https://www.postgresql.org/docs/online-resources/>

2. PostgreSQL Wiki

7. Recommended YouTube Channels

1. Postgres Conference Channel https://www.youtube.com/@postgresconf?utm_source=chatgpt.com
2. PGcasts https://www.youtube.com/@PGcasts?utm_source=chatgpt.com
<https://www.youtube.com/@PGcasts>

3. Crunchy Data https://www.youtube.com/@PGCasts?utm_source=chatgpt.com
https://www.youtube.com/@CrunchyData?utm_source=chatgpt.com

8. Recommended Courses

1. PostgreSQL Exercises https://pgexercises.com/?utm_source=chatgpt.com
2. PostgreSQL Tutorial https://pgexercises.com/?utm_source=chatgpt.com

9. Recommended Blogs

1. cybertec Blog https://www.cybertec-postgresql.com/en/postgresql-blog/?utm_source=chatgpt.com
2. Percona PostgreSQL Blog https://www.percona.com/blog/?utm_source=chatgpt.com
3. Crunchy Data Blog https://www.crunchydata.com/blog?utm_source=chatgpt.com
4. 2ndQuadrant Archive Blog https://www.2ndquadrant.com/en/blog/?utm_source=chatgpt.com

10. Recommended Books

- a. PostgreSQL Query Optimization b. The Internals of PostgreSQL c. PostgreSQL 16 Administration Cookbook d. Mastering PostgreSQL in Application Development e. SQL Performance Explained f. Designing Data Intensive Applications

11. Final notes

a. For someone with your SQL Server background, the steepest learning curve is usually MVCC + VACUUM + Linux + PostgreSQL planner behavior rather than SQL itself. Those four areas deserve extra depth throughout the year.

I would like to add also Performance topics like Indexes, statistics, Execution Plans, Parameter Sniffing, etc

b. Biggest SQL Server mindset changes

Your document correctly identifies some of these, but I would elevate them because they are usually the hardest transition areas.

SQL Server mindset	PostgreSQL reality
Instance	Cluster
TempDB	work_mem/temp files
Auto maintenance	VACUUM strategy matters
Clustered index	Heap + indexes
Parameter sniffing	Generic/custom plans
Fill factor focus	MVCC bloat focus
SQL Agent	pg_cron/Linux scheduler
AG	Streaming replication + Patroni
DMV-centric	pg_stat views + extensions

12. SQL Server mappings

SQL Server	PostgreSQL
TempDB	work_mem/temp files
SQL Agent	pg_cron/Linux scheduler
DMV	pg_stat views
AlwaysOn AG	Streaming Replication + Patroni
Resource Governor	OS + workload management
Included Columns	INCLUDE
Filtered indexes	Partial indexes
Database snapshots	no direct equivalent
Query Store	pg_stat_statements